

Technology #161

Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a closed fluid circuit that eliminates open lumens during

Engineering Technology-Transfer Dossier
analysis connections.

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a closed fluid circuit that eliminates open lumens during dialysis connections.

WHY DOES IT MATTER?

Documented problem 'flow patency loss' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 10000024: Injury [CAPSURE SP NOVUS LEAD]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2015-2026). Illustrative records: 'Heparin-Coated Tunneled Hemodialysis Catheters Improve Failure-Free Survival in ESRD.'; 'The Temporal pathological changes of venous catheter-related fibrin sheath and their implications for nursing practice.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 13 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Conduct a randomized trial comparing standard hubs versus the sealed needle-free ports; record time to first flow-patency loss over 8 wee.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Closed system technologies that isolate the fluid pathway to reduce contamination risk.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'flow patency loss' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 10000024: Injury [CAPSURE SP NOVUS LEAD]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2015-2026). Illustrative records: 'Heparin-Coated Tunneled Hemodialysis Catheters Improve Failure-Free Survival in ESRD.'; 'The Temporal pathological changes of venous catheter-related fibrin sheath and their implications for nursing practice.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a closed fluid circuit that eliminates open lumens during dialysis connections.

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a closed fluid circuit that eliminates open lumens during dialysis connections.

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: r , L .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: CONDITIONAL; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: NOT ESTABLISHED.

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$. Applicability to this invention: CONDITIONAL; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: NOT ESTABLISHED.

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.

- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'flow patency loss', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain catheter flow patency for at least 8 weeks without additional flushing	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Closed-loop system technologies that isolate the fluid pathway to reduce contamination	MODEL_HYP
DI-003	Expected effect target	Maintains catheter flow patency for ≥8 weeks by minimizing fibrin sheath formation and thrombus development due to re-catheterization	MODEL_HYP
DI-004	Domain design input pattern: required for applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering principles)	ENGINEERING_PRINCIPLES
DI-005	Domain design input pattern: mechanism applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering principles)	ENGINEERING_PRINCIPLES
DI-006	Domain design input pattern: anticipated applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering principles)	ENGINEERING_PRINCIPLES
DI-007	Domain design input pattern: device lifetime applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering principles)	ENGINEERING_PRINCIPLES
DI-008	Regulatory design input: compliance with standards	ISO 13485 (QMS) and ISO 14957 (device lifecycle management) are never inferred	STANDARDS
DI-009	Custodied observation: Complication rate	Source: https://clinicaltrials.gov/study/NCT01435746 , content_hash 1706da0f58b1f05ea7f2024b19df5dc290495cce6aa1e1	CLINICAL_DATA
DI-010	Custodied observation: Vascular access site infection rate	Source: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4168288/ , content_hash 06a25f1b216c5068f0e595a2712d5a576d47103952a3b54f	CLINICAL_DATA
DI-011	Custodied observation: Patency maintenance	Source: https://www.sciencedirect.com/science/article/pii/S1538586616000638 , content_hash 66822950c931386f20e35407383abd722652a21e5ae7bf0f	CLINICAL_DATA
DI-012	Custodied observation: Personalized medicine approach	Source: https://www.nature.com/articles/4204296 , content_hash 5ad46e7fb601c3267c7e297695d6afa01c042328f47d3d748	CLINICAL_DATA
DI-013	Custodied observation: Heparin coating	Source: https://www.sciencedirect.com/science/article/pii/S0007122600000285 , content_hash 3b6107660058d028f561901542dda30307bea6e385775f	CLINICAL_DATA

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Incorporate a needle-free, hermetically sealed port while a self-sealing membrane at the catheter hub, creating a	CONFIRMED	
DO-002	pressure regulation element realizing Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a c	CONFIRMED	
DO-003	sensing / feedback element (if closed-loop) realizing Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter h	CONFIRMED	
DO-004	termination interfaces (patient / reservoir) realizing Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub	CONFIRMED	
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal interface potential (if for non-compliance) needle-free, hermetically se	CONFIRMED	
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop) signal interface potential (if for non-compliance) incorporate a needle-free, he	CONFIRMED	
DO-007	sensing / feedback element (if closed-loop) interfaces with non-compliance interfaces (patient / reservoir) signal interface potential (if for non-compliance) off for Incorporate a ne	CONFIRMED	
DO-008	Device boundaries: patient/fluid inlet and outlet termination control/data ports required by suction/seal needle-free, hermetically sealed port	CONFIRMED	
DO-009	Flow/force/signal path geometry realizing the mechanism to incorporate a needle-free, hermetically sealed, surface finish self-sealing membrane at the cath	UNKNOWN	
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior	UNKNOWN	standard computed bounds for lumen inner diameter; verification of range
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior	UNKNOWN	standard or computed bounds for lumen length; verification of range endp
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior	UNKNOWN	standard computed bounds for surface roughness; verification of range
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior	UNKNOWN	standard computed bounds for operating pressure head; verification of
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior	UNKNOWN	standard or computed bounds for flow rate range; verification of range en
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior	UNKNOWN	standard computed bounds for viscosity of flowed medium; verification
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism in desired operating position	REDESIGNED	sensing regulates a needle-free, hermetically sealed port with a self-se
DO-017	Information path: sensing element -> decision/monitor control	CONFIRMED	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: bench flow-loop characterization across operating range — rig	UNKNOWN	future, externally build specification for DO-019
DO-019	Fixture enabling: pressure-drop mapping vs flow (identify flow transitions) — rig	UNKNOWN	future, externally build specification for DO-018

ID	Description	Status	Missing Inputs
DO-020	Fixture enabling: accelerated occlusion / fouling challenge	UNKNOWN — rig concept finalized, build type, instrumentation	fixed geometry, fixture run time, occlusion; acceptance instrumentation
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope	UNKNOWN — rig concept generated, build type, instrumentation	fixed geometry, fixture run time, occlusion; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
flow patency loss	Fails by flow patency loss: the design failure mode is a flow patency loss (Art. (n	the design failure mode is a flow patency loss (Art. (n	the design failure mode is a flow patency loss (Art. (n	the design failure mode is a flow patency loss (Art. (n
proximal occlusion by tissue ingrowth	Proximal occlusion by tissue ingrowth: the design failure mode is a proximal occlusion by tissue ingrowth (Art. (n	Proximal occlusion by tissue ingrowth: the design failure mode is a proximal occlusion by tissue ingrowth (Art. (n	Proximal occlusion by tissue ingrowth: the design failure mode is a proximal occlusion by tissue ingrowth (Art. (n	Proximal occlusion by tissue ingrowth: the design failure mode is a proximal occlusion by tissue ingrowth (Art. (n
lumen collapse under external pressure	Lumen collapse under external pressure: the design failure mode is a lumen collapse under external pressure (Art. (n	Lumen collapse under external pressure: the design failure mode is a lumen collapse under external pressure (Art. (n	Lumen collapse under external pressure: the design failure mode is a lumen collapse under external pressure (Art. (n	Lumen collapse under external pressure: the design failure mode is a lumen collapse under external pressure (Art. (n
encrustation / mineral deposition	Encrustation / mineral deposition: the design failure mode is a encrustation / mineral deposition (Art. (n	Encrustation / mineral deposition: the design failure mode is a encrustation / mineral deposition (Art. (n	Encrustation / mineral deposition: the design failure mode is a encrustation / mineral deposition (Art. (n	Encrustation / mineral deposition: the design failure mode is a encrustation / mineral deposition (Art. (n
regime transition to turbulence	Regime transition to turbulence: the design failure mode is a regime transition to turbulence (Art. (n	Regime transition to turbulence: the design failure mode is a regime transition to turbulence (Art. (n	Regime transition to turbulence: the design failure mode is a regime transition to turbulence (Art. (n	Regime transition to turbulence: the design failure mode is a regime transition to turbulence (Art. (n
check-valve seat wear	Check-valve seat wear: the design failure mode is a check-valve seat wear (Art. (n	Check-valve seat wear: the design failure mode is a check-valve seat wear (Art. (n	Check-valve seat wear: the design failure mode is a check-valve seat wear (Art. (n	Check-valve seat wear: the design failure mode is a check-valve seat wear (Art. (n

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
flow patency loss	Fails by flow patency loss: the device fails to maintain patency	design the anti-thrombotic surface	Disruptive flow with the worsted candidate failed in the test (AtrXV)
proximal occlusion by tissue ingrowth	Fails by proximal occlusion by tissue ingrowth	integrated struts with surface dissuade ingrowth	With growing struts candidate fails the test (AtrXV)
lumen collapse under external pressure	Fails by lumen collapse under external pressure	thick (less) external pressure	With the worsted candidate failed in the test (AtrXV)
encrustation / mineral deposition	Fails by encrustation / mineral deposition	surface coating	When tested in vivo, the device failed in the test (AtrXV)
regime transition to turbulence	Fails by regime transition to turbulence	altering geometry	With the worsted candidate failed in the test (AtrXV)
check-valve seat wear opening	Fails by check-valve seat wear opening	sealant and geometry	With the worsted candidate failed in the test (AtrXV)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Conduct a randomized trial comparing standard care to the device	Standard care vs device	Independent assessment	Not performed
VF-002	run killer experiment: falsification test	run killer experiment	run killer experiment	run killer experiment
VF-003	bench flow-loop characterization across operating range	bench flow-loop characterization	bench flow-loop characterization	bench flow-loop characterization
VF-004	pressure-drop mapping vs flow (identify pressure drops)	pressure-drop mapping	pressure-drop mapping	pressure-drop mapping
VF-005	delivered-flow trend below baseline in adverse conditions	delivered-flow trend	delivered-flow trend	delivered-flow trend
VF-006	kink/collapse test across duty cycle; flow to place signal across duty cycle	kink/collapse test	kink/collapse test	kink/collapse test
VF-007	pressure-flow mapping across the full operating range	pressure-flow mapping	pressure-flow mapping	pressure-flow mapping
VF-008	reverse-leakage rate measurement after accelerated cycling (measure quantity)	reverse-leakage rate measurement	reverse-leakage rate measurement	reverse-leakage rate measurement

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed	Validation not performed

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	medical-grade silicone elastomer	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter lumen inner diameter [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter lumen length [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter surface roughness [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter operating pressure head [dP Pa]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter flow rate range [Q m^3/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter viscosity of flowed medium [mu Pa.s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter occlusion growth tolerance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}

- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (0.5 GPa) test	conducting a randomised test	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-03	article for VF-001	per method	Conduct a randomized test	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-04	article for VF-002	per method	run killer experiment: NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-05	article for VF-003	per method	bench flow-loop characterization	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-06	article for VF-004	per method	pressure-drop mapping	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-07	article for VF-005	per method	delivered-flow trend	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-08	article for VF-006	per method	kink/collapse test across	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-09	article for VF-007	per method	pressure-flow mapping	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY
WP-10	article for VF-008	per method	reverse-leakage rate	NO TEST ABILITY	NO TEST ABILITY	NO TEST ABILITY

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.